Lesson 6.5: Can We Turn Off an Oncogene?

Grade Level: AP Biology (Grades 11–12)
Duration: 84 minutes (Double Period)
Unit: Cancer Biology (Extension to HHMI BioInteractive "How Does Cancer Happen?" Storyline)
Lesson Type: Inquiry-Based Data Analysis / Computational Biology

Overview

In this lesson, students investigate how computational biology can be used to study cancer therapeutics. Students learn how the transcription factor MYC regulates gene expression through interaction with MAX before analyzing authentic molecular dynamics simulation data comparing the native MYC–MAX complex with a computationally designed peptide inhibitor. Students evaluate multiple lines of evidence—including structural models, RMSD graphs, and interaction energy calculations—to determine whether the peptide represents a promising therapeutic candidate. Finally, students construct a scientific argument supported by evidence while discussing the strengths and limitations of computational modeling in biomedical research.

Learning Objectives

By the end of this lesson, students will be able to:

Students will know
- MYC is a proto-oncogene that regulates gene expression.
- MYC requires MAX to bind DNA.
- Protein-protein interactions regulate cellular function.
- Molecular dynamics simulations can predict protein behavior.

Students will be able to
- Construct models illustrating normal versus oncogenic MYC function.
- Interpret structural models of protein complexes.
- Analyze RMSD graphs to evaluate structural stability.
- Interpret interaction energy data.
- Construct an evidence-based scientific argument.
- Evaluate strengths and limitations of computational biology.

Essential Question

Can computational biology help scientists design molecules that turn off cancer-causing genes?

Driving Question

Can a designed peptide effectively inhibit MYC by competing with MAX?

NGSS Performance Expectations

HS-LS1-1
Construct an explanation based on evidence for how the structure of DNA determines the structure of proteins.
HS-LS3-1
Ask questions to clarify relationships about the role of DNA and chromosomes in coding proteins.
HS-LS1-6
Construct explanations based on evidence regarding regulation of cellular processes.

AP Biology Learning Objectives

- IST-1
- IST-2
- ENE-2
- SYI-1
- SYI-3

(Protein structure/function, gene regulation, analysis of biological data, evaluating evidence.)

Science & Engineering Practices

- Developing and Using Models

- Analyzing and Interpreting Data
- Constructing Explanations
- Engaging in Argument from Evidence

Crosscutting Concepts
- Structure and Function
- Stability and Change
- Cause and Effect

Materials
- Student worksheet
- RMSD graphs
- Interaction energy table
- MYC-MAX structure image
- MYC-peptide structure image
- Calculator (optional)
- Colored pencils

Prior Knowledge
Students should already understand:
- DNA → RNA → Protein
- Transcription factors
- Gene regulation
- Protein structure
- Mutations
- Cell cycle
- Oncogenes and tumor suppressor genes

Instructional Sequence
Part I — Engage (15 minutes)
Students read a short introduction explaining MYC and MAX. Working in pairs, students construct two models:
- Normal MYC regulation
- Oncogenic MYC overexpression

Class discussion follows.

Part II — Explore (45 minutes)
Students investigate authentic computational biology data.

Activity 1 -Compare structural models
Students examine:
- MYC-MAX complex
- MYC-peptide complex

Students identify similarities and differences.

Activity 2 - Competitive inhibition
Students explain how the peptide functions similarly to competitive inhibition.

Activity 3 - Analyze RMSD
Students learn:
- What RMSD measures
- Why lower RMSD indicates greater structural stability
- Why RMSD alone does not measure binding affinity

Students analyze simulation graphs.

Activity 4 - Analyze interaction energy
Students examine:

- Electrostatic interactions
- van der Waals interactions
- Total interaction energy

Students determine which system demonstrates stronger binding.

Part III — Explain (10 minutes)

Whole-class discussion

Students synthesize evidence from

- Protein structures
- RMSD
- Interaction energy

The teacher facilitates discussion connecting computational biology to modern drug discovery.

Part IV — Elaborate (10 minutes)

Students predict how redesigning the peptide would affect

- RMSD
- Electrostatic energy
- van der Waals energy
- Total interaction energy

Students justify predictions using evidence.

Part V — Evaluate (15 minutes)

Students evaluate the claim:

> "The designed peptide is a promising inhibitor of MYC and should continue to be developed as a targeted cancer therapy."

Students write a Claim-Evidence-Reasoning (CER) response using:

- MYC biology
- RMSD evidence
- Interaction energy evidence
- Limitations of computational modeling

Assessment

- Student models
- Group discussions
- RMSD interpretation
- Energy analysis questions

Claim-Evidence-Reasoning written response evaluated using:

- Scientific accuracy
- Use of evidence
- Biological reasoning
- Communication

Teacher Notes

This lesson introduces students to authentic computational biology while reinforcing concepts from gene regulation and cancer biology. Students analyze genuine molecular dynamics simulation data generated during a university research experience rather than simplified textbook datasets, providing exposure to modern biomedical research practices. The lesson emphasizes scientific reasoning over memorization by requiring students to integrate multiple sources of evidence before evaluating a therapeutic claim.